Readme Contents

1. What is this?

The Bcalc Recoil Calculator is an open-source firearm recoil calculator.

You can enter specific firearm and ammo data and the calculator will estimate:

- free recoil energy: total energy absorbed
- recoil velocity: speed the firearm moves backwards
- average impulse force: average force on hands/ shoulder
- perceived recoil score: to be used in recoil comparisons

An internet connection is not required once installed. No user data is transmitted.

2. Who is this for?

Anyone interested in comparing recoil related calculations of different firearm and ammunition combinations.

3. Basic Linux Installation Guide

1. Install System Dependencies

- OpenMandriva: sudo dnf install python3 python3-pip tkinter
- Vendefoul Wolf: sudo apt install python3 python3-pip python3-tk python3-venv

2. Create and Activate Python Virtual Environment

- mkdir -p ~/projects/bcalcrecoil
- cd ~/projects/bcalcrecoil
- python3 -m venv venvrecoil
- source venvrecoil/bin/activate

3. Install Python Packages

- pip install -r requirements.txt

4. Download the Files

- download the applications files and place them in ~/projects/bcalcrecoil (all .py, .csv, .png, .desktop files)

5. Run the Application

- from the activated venvrecoil environment: python3 startrecoil.py

Optional: Launch Icon

1. Copy to applications folder:

- cp bcalcrecoil.desktop ~/.local/share/applications/

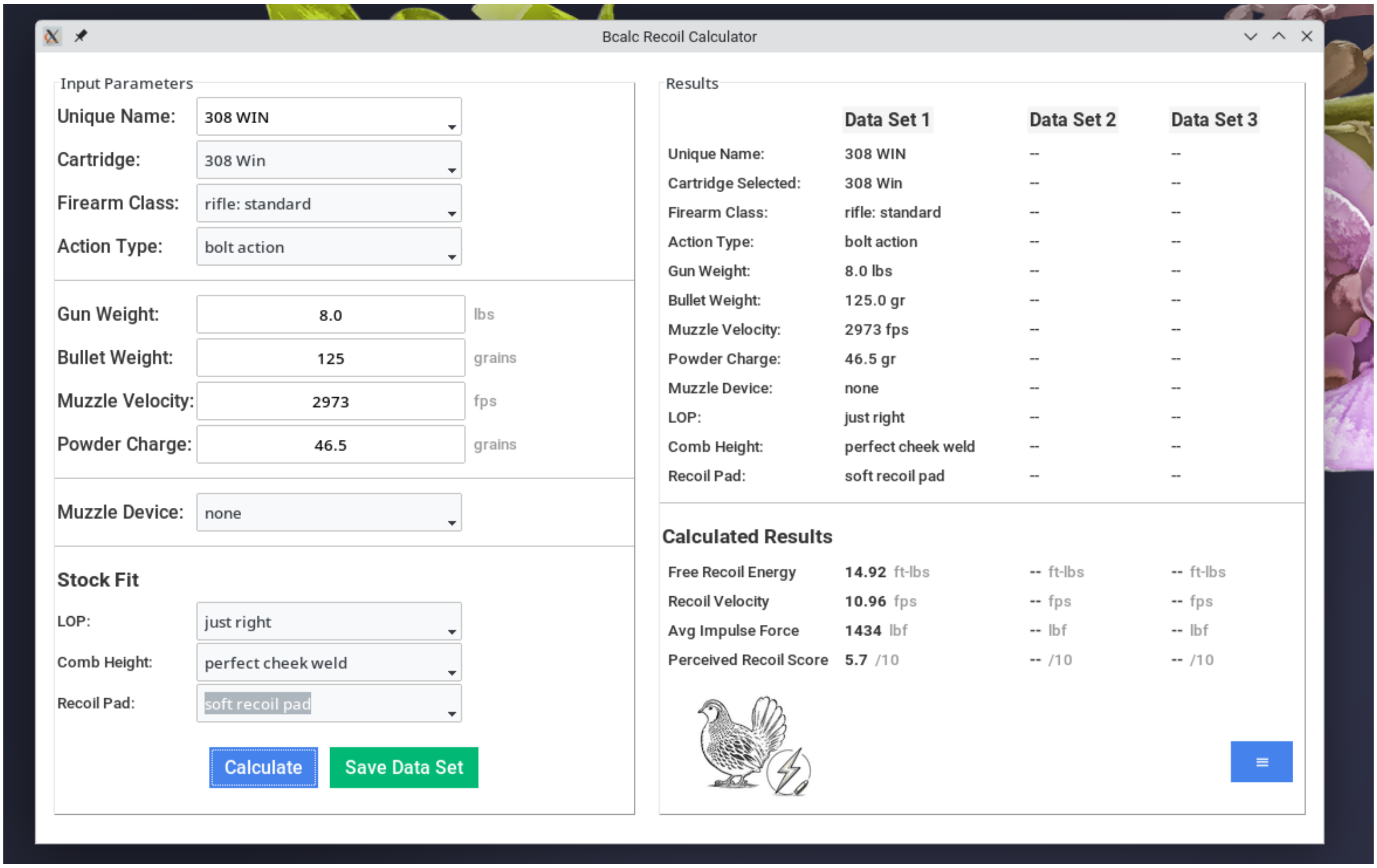
2. Copy icon

- mkdir -p ~/.local/share/icons
- cp bcalcrecoilicon.png ~/.local/share/icons/

3. Refresh desktop menu: search "Bcalc Recoil Calculator” in your app menu (should see the quail & lightning bolt image)

4. User Guide

Step 1. Enter a unique name for the data set if you want to be able to refer to it later without having to enter all the variables again. Not required however.



Step 2. Select cartridge, firearm class, and action type from the drop-down menus.

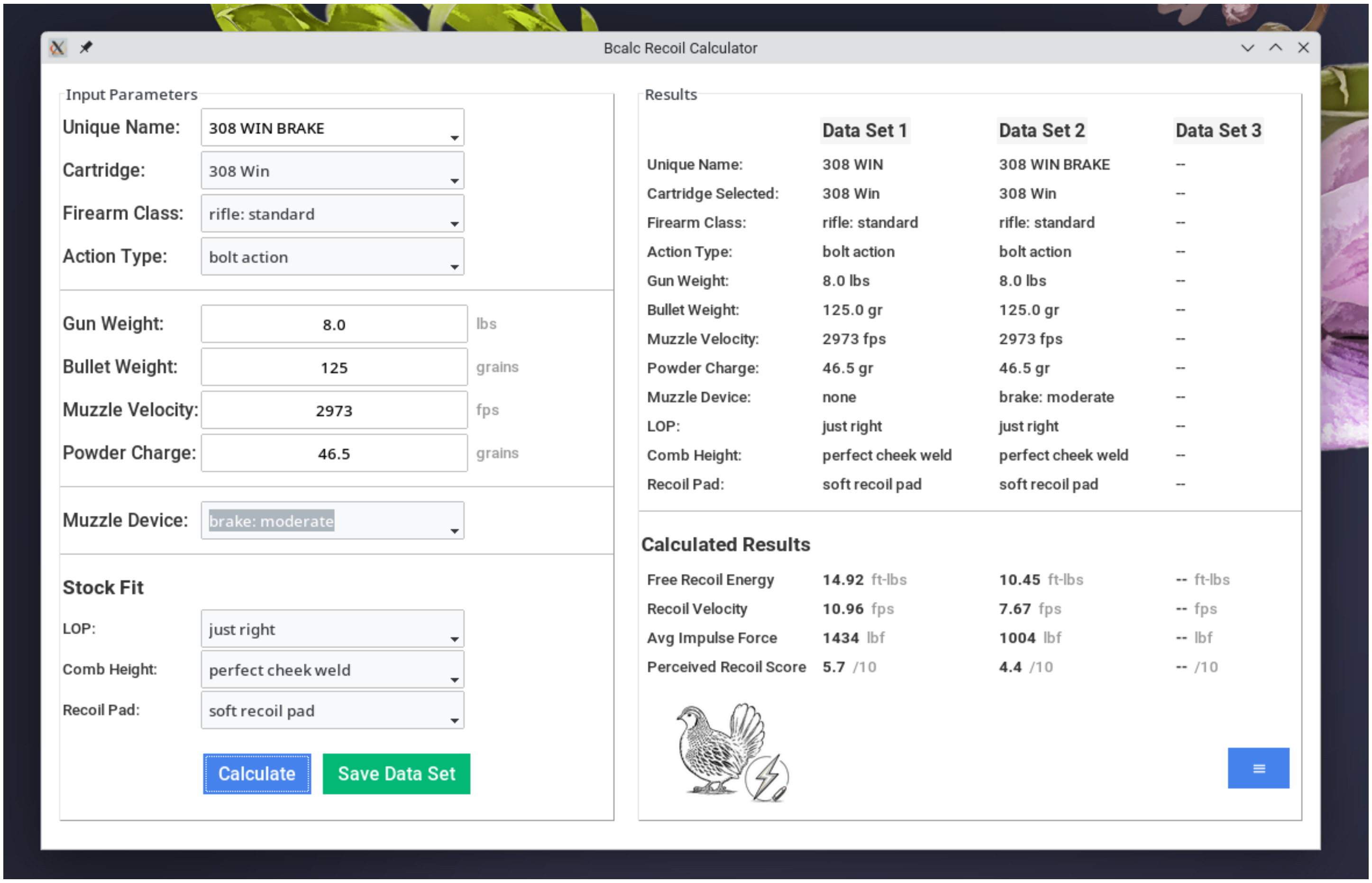
Step 3. Enter gun weight (lbs), bullet weight (grains), and muzzle velocity.

Step 4. Enter a powder charge weight (grains) if known. The calculator will estimate if left blank.

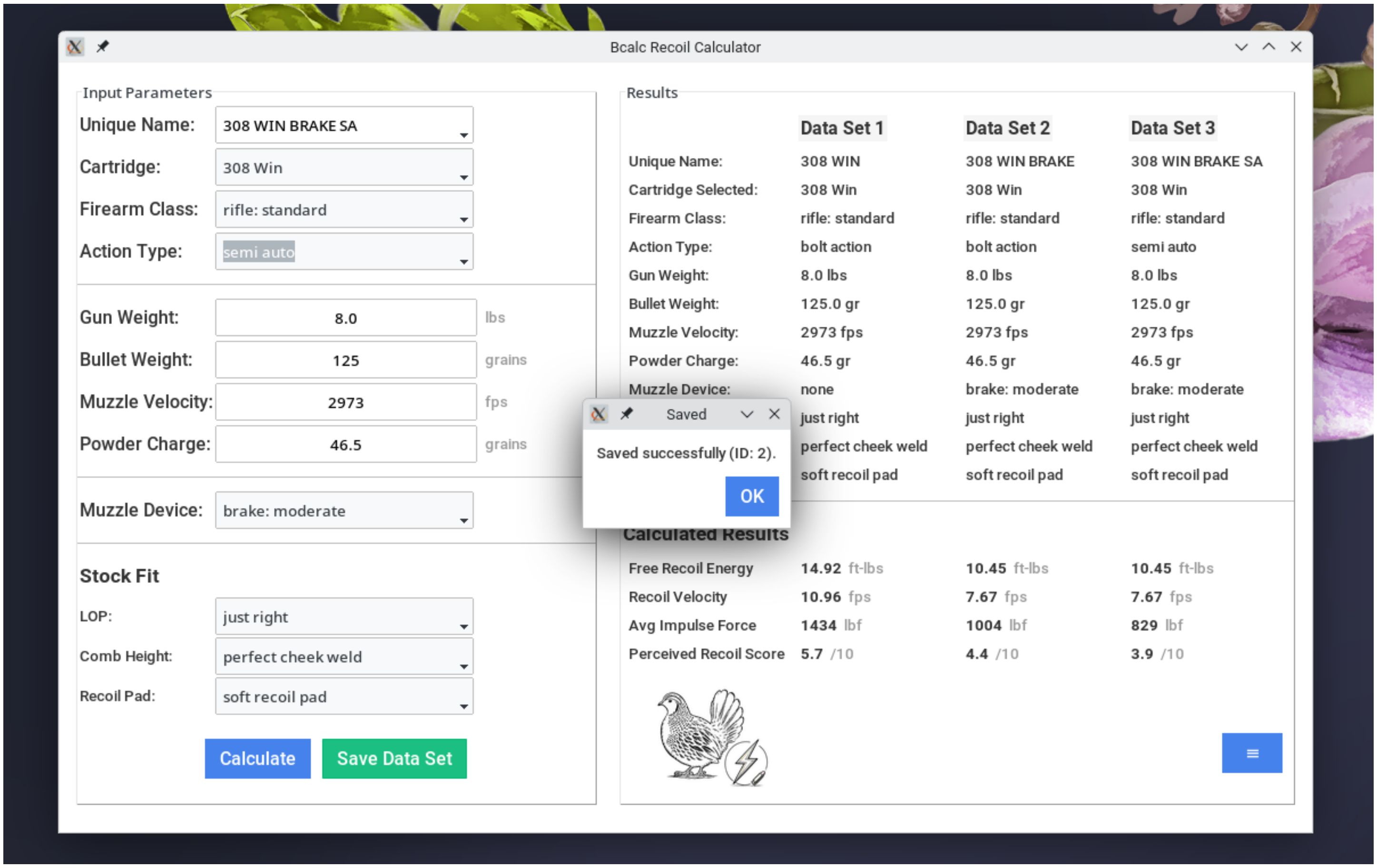
Step 5. Select muzzle device, LOP, comb height, and recoil pad from the drop-down menus.

Step 6. Press the blue Calculate button to kick off the calculator. The results show up in the right panel.

Step 7. If you wish to compare a data set, change the input parameters, and press the Calculate button. In this example, a muzzle device was added.



Step 8. Press the green Save Data Set button if you wish to refer back to a specific firearm – ammunition combination without having to reenter the data.



5. Credits & License

The Bcalc Firearm Management App was created by Brian Calc.

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